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Bixin Action in the Healing Process of Rats Mouth Wounds

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Abstract: Oral lesions that manifest as ulcer lesions are quite common and can cause discomfort to the patient. Searching for drugs to accelerate the healing of these lesions is nonstop process. Bixin is a molecule found in annatto (urucum) seeds and is considered a viable therapeutic option to treat such lesions due to its anti-inflammatory, anti-oxidant, and healing properties. Therefore, the present study aimed to evaluate the effect of the bixin solution in the ulcer healing process in the oral mucosa of rats. Ulcers were induced with punches of 0.5 cm in the middle of the dorsum of the tongue of 64 *Wistar* rats. The animals were randomly divided into 8 groups, in which 4 groups were treated with saline solution, while the other 4 were treated with the bixin solution. The animals were sacrificed in the periods 2, 7, 14, and 21 days after the beginning of the treatment. The species were histologically processed and stained with hematoxylin/eosin and picrosirius. Fibroblasts, reepithelialization, and wound contraction could be observed, as could the quantification of neutrophils, macrophages, plasma cells, lymphocytes, and mature and immature collagen. On the seventh day, the experimental group, when compared to the control group, presented a higher proliferation of fibroblasts, more advanced reepithelialization, and a higher contraction in the wounds. A reduction in the average number of neutrophils in the experimental group, when compared to the control group, could be observed in all periods (p=0.000). Up to two days, the total collagen area was higher (p=0.044) in the experimental group (4139.60±3047.51than in the control group (1564.81±918.47). The deposition of mature collagen, on the 14th day, was higher (p=0.048) in the experimental group (5802.40±3578.18) than in the control group (1737.26±1439.97). The results found in the present study indicate that the bixin solution inhibits the acute inflammatory response with a minor average number of neutrophils and accelerates reepithelialization, wound contraction and collagen maturation, thus illustrating that this solution does in fact represent an important adjuvant in the treatment of ulcers.

Keywords: *Bixa orellana*, ulcer, wound healing, herbal medicine.

INTRODUCTION

The mucosa that covers the oral cavity consists of an external layer of epithelial tissue sustained by a variable conjunctive tissue, depending on the anatomical area in which it is found. That mucosa is a barrier that protects the organism, since it prevents the penetration of any external agent that can compromise its health [1]. Physical, chemical, or biological agents can interrupt the continuity of the tissue, in turn causing a wound [1, 2].

In the oral cavity epithelium, a variety of lesions with diverse etiological factors can appear. Among the oral diseases characterized by the formation of ulcers, what stands out are recurrent herpes simplex and recurrent aphthous ulcerations [3]. Such conditions are quite disturbing to people, as they trigger an inflammatory process [1, 4].

Tissue healing can occur in the form of a regeneration with a restructuring of the tissue functional activity or through repair by reestablishing tissue homeostasis and the loss of its functional activity through fibrotic scar formation. During the evolution of the healing process, the following events frequently occur: a) the infiltration of neutrophils and macrophages, b) reepithelialization, c) angiogenesis and proliferation of fibroblasts, d) deposition of extracellular matrix, e) maturation and remodeling of collagen, and f) wound contraction [2, 5-9].

Due to the diversity of etiological factors, oral ulcer treatment is considered symptomatic. Therefore, the therapeutic resources are used to relieve pain in the area, accelerating the healing process and increasing the period without lesions [3, 4, 10, 11].

Bixa orellana L., popularly known as annatto (urucum), is commonly applied as an expectorant, a laxative, an anti–hemorrhagic, antipyretic, or anti-inflammatory drug. It is also used to treat bronchitis, tuberculosis, dyspepsia, heart conditions [12], and burns, and has secondary uses as an adjuvant in wound healing [13]. More recently, annatto was described as an antiseptic, antioxidants, diuretic, aphrodisiac, hypoglycemiant, and vermifuge drug, with a source of vitamins and with antitumoral proprieties, especially in combating oral cancer [3].

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Bixin, a carotenoid, is an isolated molecule from the seed of annatto (urucum) and is the main component of the plant's color. The carotenoids show antioxidant properties and accelerate the healing process in oral lesions [13]. Their efficiency in the healing of oral lesions has not yet been evaluated, although Pérez *et al*. (2010) [3] postulate that bixin may well be a viable option in the treatment of oral ulcers, as it presents therapeutic elements, such as anti-inflammatory, anti-oxidant, and healing properties, that are essential for tissue repair. In this sense, this study proposed to evaluate the effect of the bixin solution on the histological parameters of the healing process in the oral mucosa of rats.

MATERIALS AND METHODS

Obtaining Bixin

The *Bixa orellana* L. (urucum) fruits were collected in the Paraná Environmental Institute (IAP), in the town of Morretes, during the months of June and July of 2005. The identification was made by Dr. Gert Hatschbach and registered under protocol number 305810, in the Curitiba City Hall Botanical Museum (Brazil).

In the Federal University of Paraná (UFPR) phytochemistry laboratory, the seeds were removed from the fruits, dried in an oven at 40º, and grinded in a hammer mill. The extraction of the bixin was performed in a modified Soxhlet extractor, using a crescent polarity gradient, consisting of hexane, chloroform, and ethyl acetate, according to Costa´s technique (2007) [13]. Each extraction lasted six hours. During chloroform extraction, red-purple colored crystals, corresponding to the bixin, were obtained. Nuclear magnetic resonance was used to determine that the product of the extraction was in fact bixin [13]. The chemical structure of bixin is represented in (Fig. **1**).

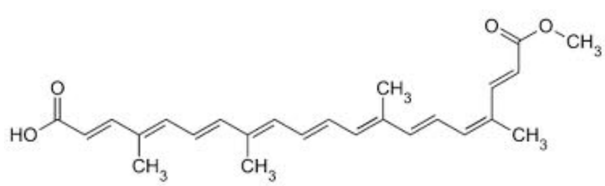


Fig. (1). Chemical structure of bixin.

After the extraction, the bixin was diluted in dimethyl sulfoxide (DMSO), at a concentration of 3mg/mL. The bixin solution concentration was chosen based on reports from Costa (2007). DMSO is a polar and aprotic compound that is commonly used in laboratories and industries in the $(CH_3)_2SO$ formula; it has a low toxicity and is a common accessory in skin compounds, easing skin absorption.

Sample

The current research was approved by the Animal Use Ethics Committee from Pontifícia Universidade Católica do Paraná (CEUA – PUCPR) under the protocol number 436, registered on May 18, 2009.

This study's sample consisted of 64 male rats, of breeds *Rattus norvegicus albinus*, *Rodentia*, *Mammalia*, and *Wistar*, approximately 45 to 50 days old, and with a corporal mass of *ca.* 200g each. The animals were fed with pelletized Nuvilab® CR-1 rations (Nuvital Nutrientes Inc., Colombo, Paraná, Brazil) and water *ad libitum*, respecting the day-night cycle, and were kept at room temperature (25±1ºC). In addition, eight rats were used as a true negative control (normal, uninjured, and untreated) for comparison in the descriptive analysis.

Inducing the Ulcers

The rats were anesthetized via intramuscular with Ketamine hydrochloride and Xylazine, at a proportion of 5mL: 1mL, applied at 0.3 mL per animal. The ulcers were induced with punches of 0.5 cm in diameter in the middle of dorsum of the rats' tongues. This technique consists of making rotating movements and placing pressure with the instrument on the area to be removed, deepening it until some form of resistance could be identified, according to that defined by Engeland *et al.* (2006) [14] and Alonso *et al.* (2008) [15]. The following three days after inducing the ulcers, the animals were treated with meperidine (2mg/kg; by intraperitoneal injection) to reduce pain and facilitate feeding.

Experimental Treatment and Control

Soon after the induction of the mechanical ulcer, the animals were randomly organized in eight groups with eight rats in each. Four groups represented the control group (for 2, 7, 14, and 21 days) and received topical applications of DMSO solution (50 µL), for one minute in all injured areas, only once a day, always at the same time. The other four experimental groups received daily topical applications of bixin solution (50µL) for one minute in all injured areas, placing a flexible stick on the ulcers for 2, 7, 14, and 21 days, respectively. The rats of each group pair (control and experimental) were sacrificed using the overdose protocol with sodium thiopental, by intraperitoneal injected, after the time periods mentioned above.

Processing and Histological Analysis

After the death of the rats, the tongues were removed with a scalpel, blade number 12, and immersed in previously identified universal collectors containing 10%buffered formalin solution for two weeks. The samples were cut in the direction of the epithelium-conjunctive tissue in such a way that two fragments were included and processed to obtain sections of a 5µm thickness. The cuts were diaphanized and stained with Hematoxylin Eosin (HE) and Picrosirius (SR). The evaluation of the histological cuts were made by two previously calibrated observers who did not know the groups.

In the qualitative analysis, lesion area; the presence, type, and inflammatory infiltration intensity; epithelialization; angiogenesis; and fibroplasia were verified by means of HE staining viewed through an optical lens (OLYMPUS® Medusa BX40) at a magnification of 200X and 400X.

In addition, the HE staining and quantitative analysis were performed through the counting of neutrophils and chronicle inflammatory cells (macrophage, lymphocytes, plasma cells) and blood vessels in all the groups and periods. For this, images were captured at 400X magnification with a Dinolite®micro camera attached to an OLYMPUS BX-50 binocular microscope, around the entire extension of the lesion. The evaluated variables were counted using the Image Tool for Windows program, version 3.0. The neutrophils were identified by their polymorphic shape nucleus. The macrophages were recognized by their irregular oval surface and an eccentric nucleus. The plasma cells were identified by their spherical and eccentric nuclei with chromatin granules, whereas the lymphocytes are cells that contain spherical nuclei and scant cytoplasm. Blood vessels were considered to be the spaces surrounded by endothelial cells [5].

The sections stained with SR were analyzed in an Olympus BX-50 microscope with a polarized Olympus® U-Pot lens and Dinolite®micro camera with a magnification of 400X. To determine the quantity of mature and immature collagen in the lesion area, images of four areas were randomly captured: two images of the superficial area and two others of the deep lesion area. The mature collagen presented an orange-red color, while the immature collagen appeared in a yellow-green color [16]. The images were analyzed using the Image Pro-Plus 4.5 morphometric program (Media Cybernetics, Silver Spring, MD), in which a mask was made to evaluate the areas in μm^2. The final values were determined from the sum of the four areas.

Statistical Analysis

The statistical was performed using the SPSS® 18.0 statistical package (SPSS Inc., Chicago, IL). The Kolmogorov-Smirnov normality test and the Levene homogeneity test were applied for all studied variables. The ANOVA-two way parametrical test and the Games-Howell or Tukey HSD test were used when recommended. In all the tests, the adopted statistically significant level was of 5% ($p<0.05$).

RESULTS

Qualitative Analysis

The histopathological analysis of the true negative control in HE showed a tongue fragment covered by squamous stratified epithelium orthokeratinized with filiform papillae. The lamina propria consisted of dense connective tissue and showed the presence of blood vessels. The submucous consisted of muscle tissue.

At day 2 of treatment in the control group, an ulcer area covered by a fibrin purulent membrane could be observed. Granulation tissue could be identified in the deep portion of the ulcer and presented intense inflammatory infiltrate with a

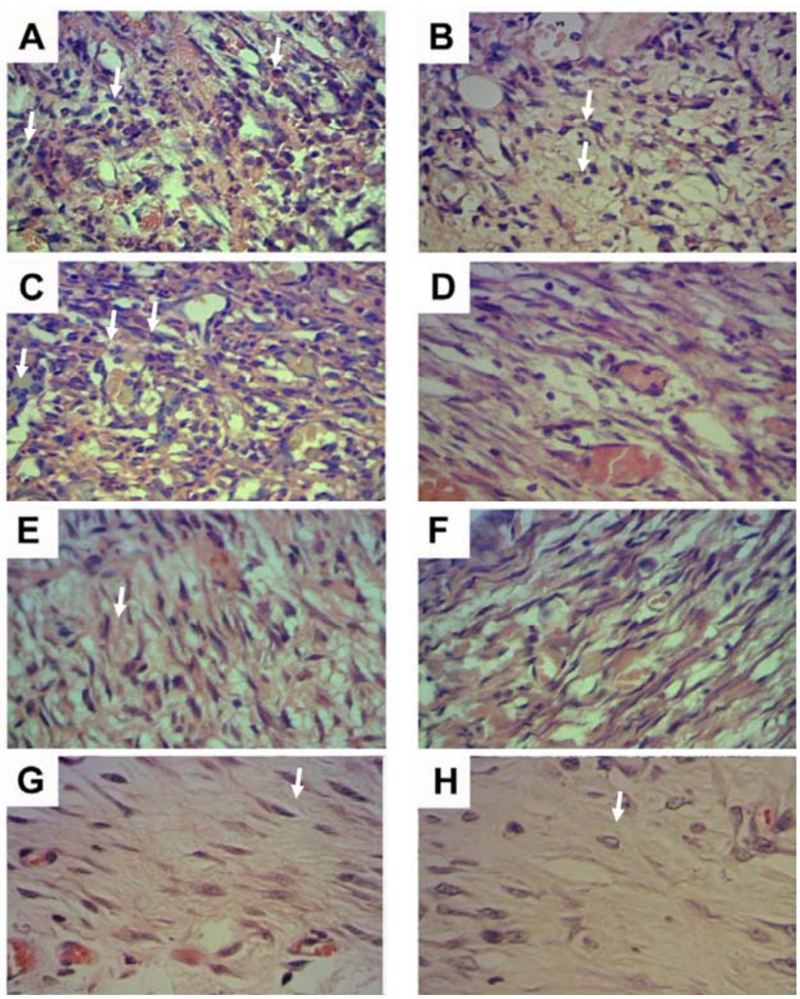


Fig. (2). Photomicrography of the injured area of the control (**A**, **C**, **E**, **G**) and experimental (**B**, **D**, **F**, **H**) groups at day 2 (**A**, **B**), day 7 (**C**,**D**), day 14 (**E**, **F**), and day 21 (**G**, **H**) (magnification 400×, HE), showing the evolution of the process over the period. Arrows: A, B -Inflammatory cells; C, D -Fibroblasts; E, F - Collagen fibers; G,H – Organizing collagen fibers.

predominance of neutrophil, severe edema, and blood vessels (Fig. **2A**). The experimental group showed a lower number of neutrophils (Fig. **2B**).

At day 07, the presence of ulceration covered by a fibrin purulent membrane could also be observed but with no complete reepithelialization of the lesion. In the granulation tissue, a reduction in the number of neutrophils and the presence of macrophages, lymphocytes, and plasma cells could be observed, increasing in the number of blood vessels and the proliferation of fibroblasts when compared to the control group at day 2. The collagen fibers were scarce and disorganized, and presented an inhomogeneous aspect (Fig. **2C**). In the experimental group, all the cases showed reepithelialization, with a reduction in the lesion area; and a discrete inflammatory infiltrate could be identified. Moreover, a larger amount of fibroblasts with a greater deposition of collagen fibers were also found (Fig. **2D**).

At day 14 of treatment, the epithelium had been completely reestablished in both groups. In the control group, the conjunctive tissue subjacent to the ulcerate area still showed a chronicle state of inflammatory infiltrate, with scattered mononuclear cells, the presence of fibroblasts, disorganized collagen fibers, and blood vessels (Fig. **2E**). In the experimental group, the organization of collagen was more evident, with thicker and cohesive fibers, showing a better organization when compared to the control group (Fig. **2F**). Once again, a minor lesion area was observed.

After 21 days of treatment, the histological features proved to be quite similar when comparing the experimental group to the control group (Fig. **2G**). Reepithelialization had been completed. In addition, in the conjunctive tissue, a discreet chronicle inflammatory infiltrate could be observed, together with presence of blood vessels, intense collagen matrix, collagen fibers of varied and organized thickness, and a reduced lesion area (Fig. **2H**).

Quantitative Analysis

The experimental group, when compared to the control group, presented a minor average of neutrophils, presenting significant statistics in two days (126.11±64.32 *versus* 232.79± 95.09; p=0.000), in seven days (95.38±46.53 *versus* 163.71± 68.01; p=0.000), in fourteen days (16.07± 13.77 *versus* 26.80± 11.52; p=0.005), and in twenty-one days (3.67±4.10 *versus* 15.48± 9.95; p=0.000) (Graph **1**).

No significant differences could be observed between the experimental and control groups as regards the average number of blood vessels (28.53±20.77 *versus* 31.44±20.25), lymphocytes (1.35±1.47 *versus* 1.97±2.04), plasma cells (1.39±1.93 *versus* 1.93±2.64), and macrophages (3.10±2.41 *versus* 4.31±4.18).

On day 2, the collagen area analysis showed a larger area of total collagen in the experimental group than in the control group (4139.60±3047.51 *versus* 1564.41±918.47; p=0.044) (Graph **2**). Concerning the immature collagen, no difference could be observed among the groups in any of the periods (Table **1**). When comparing the experimental group to the control group, a greater amount of mature collagen could be verified on day 14, (p=0.048) (Table **2**).

DISCUSSION

Bixin, a natural product, is non-toxic and contains healing, antioxidant, anti-inflammatory, and antimicrobial properties. It also contains the appropriate pharmacological features to treat mouth lesions [3, 12, 13, 17, 18]. Perez *et al.* (2010) [3] suggested that bixin should be applied in this type of therapy, although there no prior studies had been conducted that could prove such a pharmacological result in oral tissues. In this sense, the present work determines the efficiency of bixin in the treatment of induced oral ulcers.

This is the first study to evaluate the effectiveness of bixin on the oral cavity lesion repair process. The results observed revealed that such a molecule acted in the following manner: a) contraction of lesion after 7 days; b) complete reepithelialization on the seventh day; c) minor chemotaxis of neutrophils, at all the treatment periods; d) greater deposition of mature collagen at day 14.

Table 1. Average Values and Standard Deviation of Immature Collagen.

Time Group	Day 2 [a]	Day 7 [b]	Day 14 [c]	Day 21 [d]
C(μm^2)	1530.86±866.85	1724.52±1155.61	1768.40±1414.60	1413.21±593.13
E (μm^2)	3242.76±2953.81	2038.42±1068.14	2125.45±4067.74	2428.82±1856.02

Test of Games – Howell: [a] p= 0.974; [b] p= 1.000; [c] p=1.000; [d] p= 0.999.

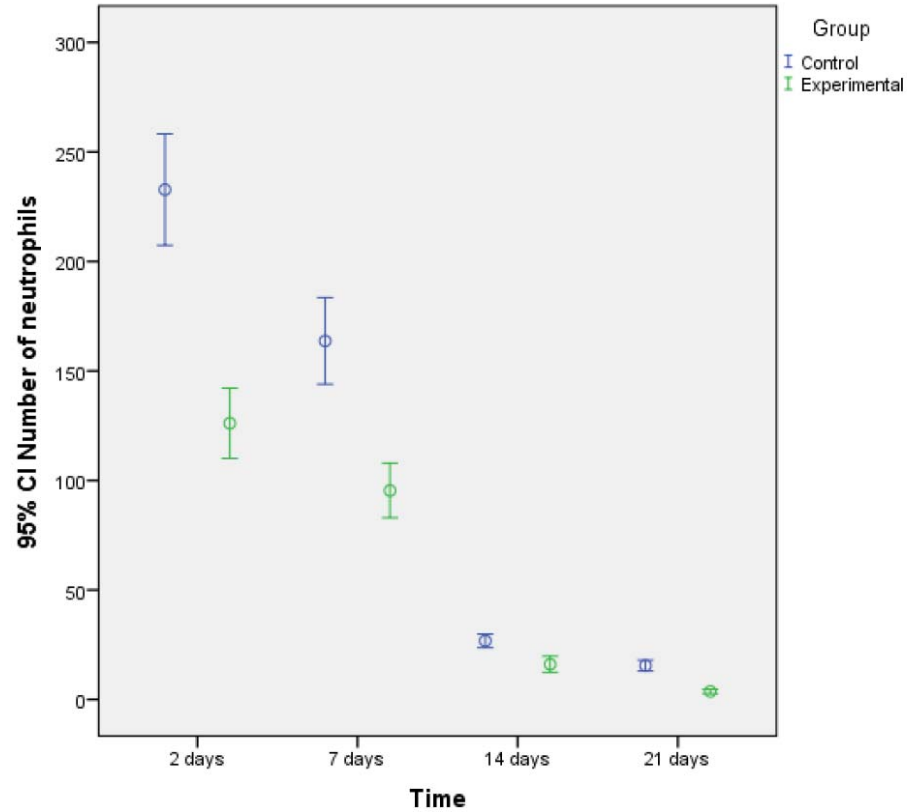
C= control; E= experimental.

Table 2. Average Values and Standard Deviation of Mature Collagen.

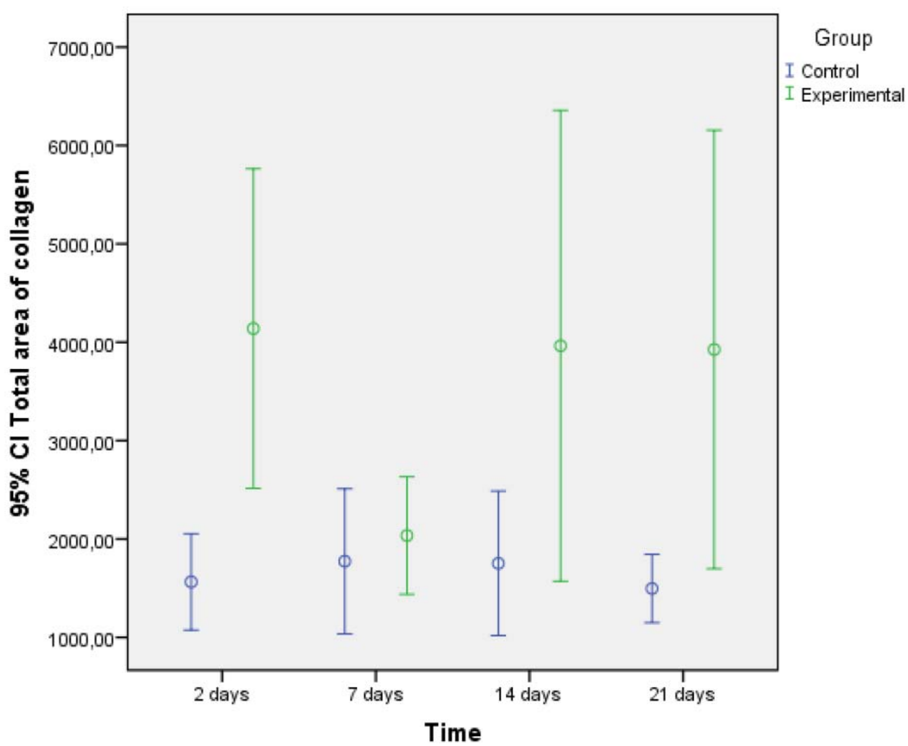
Time Group	Day 2 [a]	Day 7 [b]	Day 14 [c]	Day 21 [d]
C (μm^2)	1597.96±1026.50	1823.85±1275.26	1737.26±1439.97	1580.88±644.78
E (μm^2)	5036.43±3055.81	2033.46±1085.05	5802.70±3578.18	5423.72±4271.75

Games – HowellTest: [a] p= 0.153; [b] p= 1.000; [c] p= 0.048; [d] p =0.148.

C= control; E= experimental.



Graph (1). Average values of the number of neutrophils according to time and treatment. The experimental group, when compared to control group, presented a lower average of neutrophils in all periods, with significant statistics (in two days p=0.000; in seven days p=0.000; in fourteen days p=0.005; and in twenty-one days p=0.000). CI= Confidence interval.



Graph (2). Total area of collagen according to time and treatment. A high total area of collagen was observed in the experimental group at day 2 (p=0.044). When the other periods were analyzed, the value p was > 0.05. CI= Confidence interval.

Through the quantitative analysis of species, a reduction in the average number of neutrophils could be identified in the groups that received herbal treatment. Likewise, findings from Marchionni *et al.* (2006) [19], when using anti-inflammatory steroids in lesions induced by means of a punch on the dorsum of rats' tongues, verified a reduction in the neutrophil chemotaxis. Due to the reduction in neutrophils in the experimental group, the results found in this study suggest the need for further studies concerning the actions of bixin in neutrophil modulations. New studies should be conducted *in vitro* to analyze bixin action's impacts on neutrophils.

The main function of the neutrophils in repairing tissues is wound decontamination and protection against the invasion of possible pathogenic microorganisms in order to avoid infections, given that these can produce phagocytosis and release mediators that can act in such a process [5, 6, 8]. However, findings reported in prior literature show that, in the wound, some mediators, such as oxygen reactive species of cationic peptides, eicosanoids, and proteases, impair the tissue structures that were recently formed to repair the tissue. Elastase and other proteins secreted by the activate neutrophils cut several proteins from the extracellular matrix (MEC) and collagen IV. As MEC is responsible for sustaining the epithelial cells, as well as the migration and proliferation of keratinocytes, changes in this matrix can bring consequences to epithelial repair. Current evidence suggests that the products secreted by the neutrophils inhibit the migration and proliferation of keratinocytes [20, 21]. Dovi *et al.* [20] induced neutropenia and lesions with punches in rats. These authors observed an acceleration in the reepithelialization of lesions [20]. The present study showed that the neutrophil reduction most likely contributed to the acceleration of the reepithelialization of the wound in all periods of the experimental group. This finding corroborates with the authors cited above.

Macrophages are cells which produce cytokines that stimulate the angiogenesis and synthesis of collagen. Data from prior literature point out that a reduction in neutrophils does not necessarily imply a reduction in the macrophage needed in the tissue repair process [9, 20-22]. In the present study, despite the decrease in neutrophils, no changes could be observed in the number of macrophages, in turn allowing for a positive evolution in the repair process.

Dovi *et al.* (2003) [20], when inducing neutropenia in rats, observed no quantitative changes in the repair process in conjunctive tissues, given that the quantity of deposited collagen and the neovascularization in the neutropenic rats, presented no statistically significant differences when compared to the control group. However, the present study shows that the deposition of mature collagen is in fact accelerated in the experimental group, with statistical differences observed at day 14.

Vitamin A, a carotenoid, influences the tissue repair process, as it stimulates angiogenesis, fibroblast proliferation, collagen synthesis, wound contraction, and reepithelialization, as well as accelerates the cell turnover. These actions justify the use of cosmeceuticals as an acne treatment [23-27]. In addition to vitamin A, in the present study bixin is also proved to be a carotenoid that presents pharmacological action in the healing process by stimulating collagen maturation, wound contraction, and reepithelialization.

González *et al.* (1989) [28] and Pérez *et al.* (2004) [3] observed acceleration in a chemically induced lesion healing process in female rats that were treated for 3 days with urucum cream. These authors performed quantitative analyses of the inflammatory intensity and the clinical aspect of the wound. The bixin found in the cream was most likely the main responsible agent that led to these findings. Their results were similar; however, the present used an isolated molecule of annatto (urucum). In addition, the bixin concentration in the present study was less than that used by González *et al.* (1989) [28] and Pérez *et al.* (2004) [3], given that a lower concentration of bixin solution was used to verify the minimal effective concentration on the tissue repair process. Future studies are warranted to verify the effect of different concentrations of bixin in the overall healing process.

The total area of collagen showed statistically significant differences at day 2, at which time a superior total quantity of collagen in the experimental group, as compared to control group, could be observed. After 14 days of treatment, the mature collagen presented statistically significant differences when compared to the control group. It can be inferred that such a result stems from the fact that bixin is a carotenoid. These events are of great importance in the evolution of the healing process. Further studies are needed to evaluate the bixin effect on the collagen maturation process.

Bixin has shown favorable action as regards the tissue repair process through the modulation of the acute inflammatory response, mature collagen deposition, reepithelialization, and wound contraction. From such results, it can be concluded that bixin represents a potential adjuvant in the treatment of oral ulcers.

The figures used in the construction of the abstract were obtained from a Journal from PUCPR [29].

CONFLICT OF INTEREST

The authors confirm that this article content has no conflicts of interest.

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